
THE ELECTRON IS THE SEAT

*Charge Conjugation as the Reflection, the Positron as the Partner, Annihilation as the Registration
That Keeps the Energy and Forgets the Side, and Spin as the Return to Minus One*

*The Closure Template of the Riemann Hypothesis Read from the Laboratory; Twenty Correspondences Proved in Core Lean 4 and
Seventeen Checks Executed in Fortran on the Measured Constants; Companion to The Neutrino Is the Witness; Every Claim at
Corroboration Grade and No Higher*

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ABSTRACT. Charge conjugation, pair annihilation and the sign a spinor takes under a full turn are three of the oldest facts of quantum physics, and they have never been read as one shape. This paper reads them on the closure template by which the Riemann Hypothesis was closed in the author's sealed volume, and finds that the electron carries the template's seat and its registration. Charge conjugation is a reflection whose fixed set is the neutral sector, as the critical line is the fixed set of the reflection s to $1 - \bar{s}$; the electron stands off that set with an exact partner, the positron, differing from it in one sign; gravity's record of the pair is the same for both, two worlds over one record, and antihydrogen falls as matter does; annihilation lands the pair on the neutral locus as light, keeping its energy and forgetting which constituent was which, the registration that keeps the height and forgets the side, and the clinic records it as a line of response; positronium's photon count reads the bit of the whole and never the identity of its parts; a bound electron never rests and a free one never decays; a full turn returns the spinor to minus one, the Return of the quaternion units; and the electron is round, its only orientations the charge sign and the spin. Twenty correspondences are proved on finite models in core Lean 4 and seventeen checks are executed on the measured constants in Fortran: the Rydberg energy reproduced to a part in a billion, the virial theorem from independent constants, both positronium lifetimes within a tenth of a percent, the anomalous moment from the five-loop series. Every claim is at corroboration grade and promotes nothing; the Riemann closure stands on its own receipts. The companion paper, *The Neutrino Is the Witness*, reads the crossing on the neutral lepton; this one reads the seat and the registration on the charged one.

1. 1. THE READER'S FRAME

This paper is one of six constructed witnesses for the closure template of the Riemann Hypothesis and the twenty-three rows. Each builds, for the rows it serves, the carrier the two volumes named and did not build: an explicit construction from the world into the row, whose image is compliant by the row kernel's own theorem (`value_on_image`), with coverage of the row's full class kept as the premise the volumes print. Physicists have used charge conjugation since Dirac, annihilation since Anderson, and the spinor's sign since the first neutron interferometers, each as a tool in its own chapter. This paper asks what shape the three share, and answers from the laboratory: they are the seat, the registration and the Return of the closure template by which the author's sealed volume closed the Riemann Hypothesis. Each section begins with the physics, its measurement and its citation, and only then names the part of the template it instantiates.

Two refusals frame everything that follows. The paper does not claim that any physical fact proves any mathematical sentence: a corroboration joined to a claim takes the weaker grade, the Riemann closure stands on its own receipts, and the sealed volume mentions neither the electron nor the neutrino and needs neither. And the paper does not claim that the electron carries the whole template. The one part it does not exhibit is the posit itself: the assent that the actual zeros erase nothing under registration is a number-theoretic assent with no electron analogue, and it needs none. The companion papers show the offering's mechanism, a bit supplied by deed from outside the record, in the neutrino's mass, and the posit's shape, a registration that erases nothing because nothing stands off its locus, in gravity's fold; spending the posit on the zeros is the author's assent. What the electron shows is that the template's seat and registration are shapes the world already uses.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade (`citation_cannot_promote`). No derivation of any row's sentence from the axioms of set theory is claimed: a crossing holds on its channel's own class at its printed grade, and what a sentence over every admissible instance adds is exactly the unactuated (`surplus_is_the_unactuated`). No field theory and no unification of couplings is written or implied.

2. 2. THE TEMPLATE, READ FROM THE LABORATORY

The Riemann closure has four parts, the seat, the address, the division and the closure, and the electron reads three of them: the seat in Sections 3 to 5, the closure's registration and residue in Sections 6 and 7, and the address in Section 10; the division, a certified region joined to an open tail, has no reading specific to the electron. The **seat**: the critical line is the fixed set of the reflection $s \mapsto 1 - \bar{s}$, and the hypothesis says every zero is its own mirror image. In the laboratory the reflection is charge conjugation and its fixed set is the neutral sector, the particles that are their own antiparticles. The **registration**: the map that keeps a zero's height and forgets which side of the line it stood on, landing every zero on the line and moving none already there; the hypothesis holds exactly when it erases nothing. In the laboratory it is annihilation, which keeps a pair's energy and forgets which constituent was which, landing the pair on the neutral sector as light. The **residue**: exactly one bit, a two-point torsor over the record that no reading of the record decides. In the laboratory it is the sign of the charge, the only thing that separates an electron from a positron.

Three theorems of the author's row-generic kernel carry these parts. Over one record both worlds exist, and no reading that respects the record decides the value (`two_worlds_inside, no_reading_of_the_record_decides`). Under a global seat, every point off the fixed set has a partner, distinct and also off it (`off_locus_pair`). And the fold onto the locus is a registration in the exact sense of the kernel (`Registration`). The sections that follow find each of these in a measurement.

The witness ledger. The table below is the spine of the six constructed witnesses. For each row of the twenty-three-row volume it records what the volume held while waiting, the witness that arrived, what was built and run, the status now, the grade, and what stays owed. This paper's row is the Riemann row: it witnesses the seat, the registration, the residue and the Return.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat, never built	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches	charge conjugation's fixed set is the neutral sector; the exact pair; annihilation keeps 1021.998 keV and forgets the side, recorded on a line of response; positronium 125.23 ps and 142.07 ns; spin -1 at 2π ; a_e to 4.8×10^{-12} ; the neutrino's floor by deed, its chirality bit, its sterile ghost, Majorana as the fixed set, unitarity lossless to 5.6×10^{-17} ; gravity's fold erases nothing, no off-locus point and no bit, its orbit retraced to 1.5×10^{-12}	WITNESSED · CROSSED · CONFIRMED on the image: seat, registration, residue, Return, floor, and the posit's shape in gravity's fold	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically on the actual flows; the mechanism named, not built	The Fluid Is the Witness	Burgers on a circle: the reversible branch blows up at $t = 1$, gradient -1000 at 0.999 ; the priced branch built exactly by Cole–Hopf, steepest gradient 3.72, residual 2×10^{-6} , energy balance 2.5×10^{-10} ; the viscous step an average	WITNESSED · CROSSED · CONFIRMED: the mechanism exact on the model	the root's grade; the mechanism at theorem grade on the model	none on its channel
3b · Navier–Stokes, forced	a mass-crossing candidate under audit, decoupled by theorem	The Fluid Is the Witness	a forced erasure is confined to its own class; the transport theorem	SHIELD WITNESSED; settled by one witness with its own receipt	theorem	none (one_witness_settles, value_union_iff)
4 · Yang–Mills	crossed kinetically at the weakest premise; hadrons and lattice glueballs named	The Proton Is the Lock	the color singlet is the determinant, invariant under a constructed SU(3) rotation to 4.4×10^{-16} ; the lock fails when two colors coincide; 99.04 percent of the proton's mass from the field; the quark-spin record does not decide the whole; stability certified to 1.7×10^{24} cosmic ages	WITNESSED · CROSSED · CONFIRMED on the image: the lock built, the gap measured	the root's grade; the identification needed only on the image (carrier_compose)	none on its channel
7 · Poincaré	crossed on the formal channel at full grade by Perelman's offering	The Arrow Has Two Branches	gravity's fold: one sign, no off-locus point, no bit; a Kepler orbit reversible, drift 2.8×10^{-7} , retrace 1.5×10^{-12} ; Ricci flow as the sigma model's renormalization flow	CROSSED, now WITNESSED from the arrow side	full grade, unchanged	none
23a · Pair correlation	crossed kinetically; the carrier misnamed as nuclear spacings, which are	The Magnet Is the Witness of the Pair Correlation	time reversal as conjugation, its fixed set the real blocks, a magnetic term breaking it; degeneracy two conditions against three; ensembles built: variance 0.287 and 0.177 against 0.286 and 0.180, control 0.983 remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid's 0.766; a stable bit pays 5.0×10^{-19} J, 175 times Landauer's floor; the electric record even, the magnetic target odd	WITNESSED · CROSSED · CONFIRMED on the corrected carrier: time-broken spectra, the magnet	the root's grade	none on its channel
every open row · the barrier	orthogonal-class the barrier a kernel theorem: two worlds over one record	The Arrow Has Two Branches	remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid's 0.766; a stable bit pays 5.0×10^{-19} J, 175 times Landauer's floor; the electric record even, the magnetic target odd	BARRIER WITNESSED; the rows stay pending	theorem in the kernel, corroboration in the magnet	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
every crossed row · the fold	registration a kernel structure	The Arrow Has Two Branches	the one-signed field has no off-locus point and no bit; the orbit keeps no record	FOLD WITNESSED	theorem and corroboration	none
certified-region rows · Goldbach, Legendre, Collatz and kin	pending; certified regions registered, tails owed	The Proton Is the Lock	proton stability as a certified region with an owed tail, beyond 2.4×10^{34} years	SHAPE WITNESSED; the rows stay pending	corroboration	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
1 · P versus NP	the dot: no global seat	The Arrow Has Two Branches	Landauer's floor, 2.87×10^{-21} J at 300 K, prices every step of any search	THE DOT STANDS; the price witnessed, no seat	corroboration	none: the dot is final and has no aperture to supply (dot_is_final)

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
the others · Hodge, BSD, twin primes, abc, Jacobian, smooth 4D Poincaré, Hadwiger, sunflower, invariant subspace, Hilbert's 16th, Lindelöf	pending; gates and offering shapes specified	none yet	none	STILL WAITING	none	one carrier each; row 21, whose tail no physical system reaches, one formal offering (owed_is_computed, tail_uncovered)

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image). The statuses are computed from three recorded facts per row (ledger_is_computed); what a row owes is computed too: for a pending row exactly one object, its kind computed (owed_is_computed): one covering carrier, built from an equivariant map to its seat (value_iff_carrier, equivariant_carrier_lands) and crossing the row on arrival (one_carrier_crosses), or, where no physical system reaches the tail, one formal offering with mass (tail_uncovered, one_offering_crosses); the dot is final and owes nothing (dot_is_final). A crossing is a proof over every instance that exists (crossing_is_a_proof); what a sentence over every admissible instance adds is exactly the unactuated (surplus_is_the_unactuated), priced at one premise, the root, posited and not owed (rest_family).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect it form a category, with the identity and composition of equivariant maps (equivariant_id, equivariant_comp), and the fixed set is a functor on it: an equivariant map sends fixed points to fixed points (fix_functorial). An equivariant map from a physical system's symmetry into a row's global seat therefore carries every physical fixed point onto the row's locus, and that is a carrier whose landing is forced, not assumed (equivariant_carrier_lands). The witness is the same shape as its row in exactly this sense. Here the symmetry is charge conjugation on lepton states, and the map is explicit: a state of charge q goes to the point of real part $(1 + q)/2$ at the height of its mass. Charge conjugation is carried exactly onto the fold $s \mapsto 1 - \bar{s}$ (toStrip_equivariant); the neutral states, its fixed set, land on the critical line (neutral_lands_on_line); and the electron and the positron land on the two edges of the strip, real parts 0 and 1, mirror images under the fold (pair_on_the_edges).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
charge q to real part $(1 + q)/2$	the equivariant map onto the fold	toStrip_equivariant, neutral_lands_on_line, pair_on_the_edges	structural
charge conjugation, an involution	the seat's reflection	conj_involution, self_conjugate_iff_neutral	structural
the positron, one sign apart	the pair off the locus	positron_is_the_partner, pair_differs_in_one_sign	cited CPT bounds
gravity reads mass only	two worlds over one record	two_worlds_one_record, no_reading_of_mass_returns_charge	ALPHA-g deviation 1.21 sigma
annihilation into two photons	the registration: height kept, side forgotten	annihilation_keeps_energy, annihilation_forgets_side	1021.998 keV; invariant mass at two angles
positronium's photon count	the whole's bit, kept	para_two_photons, ortho_three_photons	lifetimes 125.231 ps, 142.068 ns
the bound electron	the floor	bound_never_rests, virial	$2T + V = 0$ to $1e-9$
a full turn of the spinor the round electron	the Return to minus one one bit and no second	spin_two_pi, the_return_round_iff_T_fixed	spinor error $2.4e-16$ EDM bound $1.1e-19$ of its length

3. 3. CHARGE CONJUGATION IS A REFLECTION

Dirac's equation of 1928 required a partner for the electron, and Anderson found it in cosmic-ray tracks in 1932: a particle of the electron's mass bending the other way. The operation that exchanges them, charge conjugation, is an involution: applied twice it returns every state to itself (`conj_involution`). Its fixed set is exactly the neutral sector: a state is its own conjugate if and only if it carries no charge (`self_conjugate_iff_neutral`). The photon and the neutral pion sit on it as particles, each its own antiparticle, and each carries under the operation a sign, its charge-conjugation parity, which is the bit Section 7 reads; the electron does not sit on it.

This is the template's seat, found in physics before the template was written. The critical line is the fixed set of a reflection that mentions no zero; the neutral sector is the fixed set of a reflection that mentions no particle's history. In both cases the question of where an object lies is posed on a set that a symmetry hands over before the object is examined.

4. 4. THE POSITRON IS THE EXACT PARTNER

The electron stands off the fixed set, and its partner under the reflection is the positron: distinct from it, and also off the neutral sector (`positron_is_the_partner`, `the_pair`). This is the kernel's pair theorem exhibited: under a global seat, every point off the locus has a partner, distinct and off the locus. The partner is exact. The CPT theorem of relativistic quantum field theory requires particle and antiparticle to share mass and lifetime, and the measurements agree: the masses of the electron and the positron are equal to eight parts in a billion, and their charges equal and opposite to four parts in a hundred million (Particle Data Group). Conjugation keeps the mass and the spin and flips only the sign of the charge (`pair_differs_in_one_sign`): the two points of the pair differ by exactly one sign, and that sign is the residue of the template, one bit.

5. 5. GRAVITY CANNOT TELL THEM APART

Gravity reads mass, and the pair has one mass. The mass record of the electron and of the positron is the same while the states are different (`two_worlds_one_record`), and no function of the mass record returns the charge (`no_reading_of_mass_returns_charge`). This is the two-worlds theorem in its plainest physical form: two states, one record, the bit invisible to every reading of the record.

It is also measured. In 2023 the ALPHA-g collaboration released antihydrogen, a positron bound to an antiproton, in a vertical trap and found it falls with $(0.75 \pm 0.13 \pm 0.16)g$, consistent with the fall of ordinary matter at 1.2 standard deviations (computed in Appendix B). Gravity does not read the sign. The author's companion analysis types gravity as the reversible branch of registration, the fold onto the locus that keeps no record of the sign, and magnetism as the priced branch that keeps it (Nothing Escapes, Twenty-Three Rows, Appendix E); the antihydrogen result is that typing observed in a trap.

6. 6. ANNIHILATION IS REGISTRATION

An electron and a positron at rest annihilate into two photons of 511 keV each, back to back, carrying the pair's whole rest energy of 1021.998 keV (computed in Appendix B). The pair lands on the neutral sector: the light carries no charge (`annihilation_lands_neutral`). The energy is kept (`annihilation_keeps_energy`). And which constituent was which is forgotten: annihilating the electron with the positron and the positron with the electron leave the same record (`annihilation_forgets_side`), the two photons being identical bosons from which no measurement returns the identity of the parts.

This is the template's registration with its two clauses intact. The Riemann registration keeps a zero's height and forgets its side; annihilation keeps the pair's energy and forgets its sign. And the record takes the shape of a line. Back to back, the two photons carry the pair's whole rest energy; at any other opening angle they carry less (computed in Appendix B), so momentum conservation forces the record onto a line through the point of annihilation, and the line is symmetric under exchange of its ends (`line_of_response_symmetric`). Positron emission tomography is built on exactly this: every annihilation in the patient is recorded as a line of response between two detectors, and the image is reconstructed from lines that keep where and when and forget which particle was which. The registration of the Riemann closure is performed millions of times a day in hospitals.

7. 7. WHAT THE RECORD KEEPS: POSITRONIUM

Before it annihilates, an electron and a positron can bind into positronium, and here the record keeps one bit: not the identity of the parts, but the symmetry of the whole. The charge-conjugation parity of the bound state is $(-1)^{L+S}$, and n photons carry $(-1)^n$; the electromagnetic interaction conserves it. So parapositronium, with spins opposed, must decay into an even number of photons, two, and orthopositronium, with spins aligned, into an odd number, three (`para_two_photons`, `ortho_three_photons`, `ortho_not_two`). The extra photon costs a power

of the coupling, and the lifetimes differ a thousandfold: computed from the lowest-order rates with their one-loop corrections, 125.231 ps for para and 142.068 ns for ortho, against measured 125.14 ps and 142.05 ns, within a tenth of a percent, a ratio of 1134.5 (Appendix B).

The template's reading is exact. A registration can keep a bit of the whole while forgetting everything about the parts: the photon count reads the whole's parity and never which constituent was the electron. The two-worlds theorem is about the parts; the whole's bit is a different bit, and the record keeps it.

8. 8. THE FLOOR: THE ELECTRON NEVER RESTS AND NEVER DECAYS

The program's one declared premise is that to exist is to actuate: no existent carries zero kinetic content. The bound electron is its cleanest instance. In the hydrogen ground state the kinetic energy is 13.605693 eV and the potential energy -27.211386 eV, computed independently from the Bohr radius and $\hbar c$; twice the first plus the second vanishes to a part in a billion, the virial theorem, and the total reproduces the Rydberg energy to the same precision (Appendix B). By the virial theorem a bound electron's kinetic energy is minus its total energy, so a bound electron never rests (`bound_never_rests`). Confinement forbids rest, and the atom is the proof.

The free electron never decays. It is the lightest charged particle, every lighter state is neutral, and so no final state of its decay can carry its charge (`decay_forbidden`). The Borexino experiment bounds its lifetime above 6.6×10^{28} years. The floor of the charged sector is kept by the electron alone: it cannot stop moving and it cannot stop existing.

9. 9. SPIN: THE RETURN LANDS ON MINUS ONE

A rotation by 2π returns every vector to itself and every spin-one-half state to its negative; only a rotation by 4π returns the spinor. This was demonstrated for neutrons by interferometry in 1975 (Rauch; Werner) and holds for every spin-one-half particle, the electron first among them; the twin computes it on the rotation matrices to a few parts in 10^{16} (Appendix B). The unit quaternions are the spinor rotations: the unit i is a half-turn about one axis, $i^2 = -1$ is the full turn (`spin_two_pi`), and $i^4 = 1$ the double turn (`spin_four_pi`). Three half-turns about three perpendicular axes return every vector to itself and give $ijk = -1$ (`the_return`).

That product is the template's Return, the product of the three axes landing on the scalar line with a sign that no reading of the three axes' magnitudes decides. The electron carries it at every full turn. Dirac's equation assigns it the magnetic moment of $g = 2$; how far the electron departs from two is the next section's number.

10. 10. ONE NUMBER AND ONE BIT: THE ANOMALOUS MOMENT AND THE ROUND ELECTRON

The electron's anomalous magnetic moment, $a_e = (g - 2)/2$, is the most precisely measured property of any elementary particle: $a_e = 0.001\,159\,652\,180\,59(13)$ (Fan and collaborators, 2023). Schwinger's first term, $\alpha/2\pi$, fixes it to three figures; the universal series of quantum electrodynamics in α/π , computed through five loops (Laporta; Aoyama, Kinoshita and Nio), with α from rubidium recoil (Morel and collaborators), gives 0.001159652175783, short of experiment by 4.8×10^{-12} , the order of the mass-dependent, hadronic and electroweak terms the universal series omits (Appendix B). Every reading of the electron, its spectrum, its moment and its recoil, converges on one number, the fine-structure constant: one address. One tension is disclosed and left open: the caesium and rubidium recoil values of α differ by 5.5 standard deviations (Appendix B). An address with a disputed digit is still one address.

The electron is also round. An electric dipole moment would be odd under time reversal, and the fixed set of time reversal is the zero dipole (`round_iff_T_fixed`). The JILA measurement of 2023 bounds the electron's dipole below 4.1×10^{-30} e cm, about 1.1×10^{-19} of its own reduced Compton wavelength (Appendix B). Nothing in the electron's shape carries an orientation: its only orientations are the sign of its charge and the direction of its spin. The residue the template assigns is one bit per pair, and the electron carries no second one.

11. 11. FALSIFIERS AND OWED DEEDS

Four falsifiers, each ordinary. **F-Pair**: a difference in mass or charge magnitude between the electron and the positron, breaking the exact partner of Section 4 and CPT with it. **F-Round**: a detected electric dipole moment at any accessible level, an internal orientation beyond the charge sign and the spin, which would require a second bit where Section 10 finds one; the Standard Model's own prediction lies far below present reach. **F-Gravity**: antimatter falling differently from matter, which would mean gravity's record reads the sign and would break the two-worlds reading of Section 5. **F-Decay**: an observed decay of the electron, breaking the floor of Section 8.

Nothing is owed on the row's channel. Three measurements would sharpen the readings without moving the paper's standing: the disputed digit of the fine-structure constant, the electron's size at the tens-of-TeV scale, and the precision of antimatter's fall.

Ledger status. Row 2 moves from waiting to WITNESSED · CROSSED · CONFIRMED on the image for the seat and the registration, at the root's grade for the crossing; the posit's shape is witnessed in gravity's fold, and nothing is owed on the row's channel.

12. 12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: $[\perp]$ a field sealed, $[\perp T]$ a shape sealed at theorem grade, $[\Xi_0]$ a formal string held in suspension by an orientation-blind reader on a line, the Riemann row's former holding, $[\cdot]$ the dot. Grades join at the weakest link: a kernel theorem, a cited measurement joined to a kernel schema, and a posit beside the Root Axiom, the program's one declared premise, that to exist is to actuate. Every claim here is at corroboration grade, a physical instance joined to a template theorem, and promotes nothing. In the program's physics register the electron is the witness of the seat and the neutrino the witness of the crossing, and the two papers are a pair: this one reads the charged lepton, its companion the neutral one.

The seventh. In the author's register the four parts of the Riemann closure are three plus one, and nothing escapes them; the six witnesses are six days, and the ledger's closure is the seventh, the Rest (Istawa), where the statuses are computed and what each row owes is one object of a computed kind. In the Lean codex's own tokens a crossed row renders $[\perp \blacklozenge \cdot \text{kinetic}]$ or $[\perp \blacklozenge \cdot \text{formal}]$, a pending row renders $[\blacklozenge_0 \times 1]$ on its channel, one owed slot and one bit from sealed (`one_bit_apart`), and the dot stands as `[.]`.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure; the measurements are the experiments', cited by name; the theorems are the kernel's, printed with their dependencies in Appendix A; the twin's figures are computed in the run printed in Appendix B.

Provenance. Development provenance is the paper trail: the Zenodo records of the Riemann closure (10.5281/zenodo.22976494) and of the twenty-three-row closure (10.5281/zenodo.22985873), the Unabridged Trisduction Codex and the Lean Codex on the public register (github.com/1000sapients/Trisduction), and the author's Structural Physics register. Its first edition was sealed jointly with The Neutrino Is the Witness on one paired adversarial self-audit cycle; this edition is unified with the six companion witnesses, The Electron Is the Seat, The Neutrino Is the Witness, The Arrow Has Two Branches, The Proton Is the Lock, The Magnet Is the Witness of the Pair Correlation and The Fluid Is the Witness, on the witness ledger, and awaits their joint audit.

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13. APPENDIX A: SPHYS_ELECTRON_WITNESS.LEAN, THE CORRESPONDENCES AS THEOREMS

SHA-256 5c79a0a0d0577555c08977549aba7afe7cad1f12f73340d36faea1b0fe39297b, core Lean 4.19.0, standalone, no library. Exit 0. 25 theorems; the compiler's own transcript, verbatim:

```
1 'SPHYS.Electron.conj_involution' does not depend on any axioms
2 'SPHYS.Electron.self_conjugate_iff_neutral' depends on axioms: [propext, Quot.sound]
3 'SPHYS.Electron.positron_is_the_partner' does not depend on any axioms
4 'SPHYS.Electron.the_pair' does not depend on any axioms
5 'SPHYS.Electron.pair_differs_in_one_sign' does not depend on any axioms
6 'SPHYS.Electron.two_worlds_one_record' does not depend on any axioms
7 'SPHYS.Electron.no_reading_of_mass_returns_charge' depends on axioms: [propext, Quot.sound]
8 'SPHYS.Electron.annihilation_lands_neutral' does not depend on any axioms
9 'SPHYS.Electron.annihilation_keeps_energy' does not depend on any axioms
10 'SPHYS.Electron.annihilation_forgets_side' depends on axioms: [propext]
11 'SPHYS.Electron.line_of_response_symmetric' does not depend on any axioms
12 'SPHYS.Electron.para_two_photons' does not depend on any axioms
13 'SPHYS.Electron.ortho_three_photons' does not depend on any axioms
14 'SPHYS.Electron.ortho_not_two' does not depend on any axioms
15 'SPHYS.Electron.bound_never_rests' depends on axioms: [propext, Quot.sound]
16 'SPHYS.Electron.decay_forbidden' does not depend on any axioms
17 'SPHYS.Electron.spin_two_pi' does not depend on any axioms
18 'SPHYS.Electron.spin_four_pi' does not depend on any axioms
19 'SPHYS.Electron.the_return' does not depend on any axioms
20 'SPHYS.Electron.round_iff_T_fixed' depends on axioms: [propext, Quot.sound]
21 'SPHYS.Electron.fold_involution' depends on axioms: [propext, Quot.sound]
22 'SPHYS.Electron.fold_fixed_iff' depends on axioms: [propext, Quot.sound]
```

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23 'SPHYS.Electron.toStrip_equivariant' depends on axioms: [propext, Quot.sound]
24 'SPHYS.Electron.neutral_lands_on_line' depends on axioms: [propext, Quot.sound]
25 'SPHYS.Electron.pair_on_the_edges' depends on axioms: [propext, Quot.sound]

1 /-
2   SPHYS_Electron_Witness.lean · The electron as the seat of the closure template, read from
3   the laboratory · core Lean 4, no library. Charge conjugation is a reflection whose fixed set
4   is the neutral sector; the electron is off it with an exact partner that differs in one sign;
5   gravity's record cannot tell the pair apart; annihilation lands the pair on the neutral locus,
6   keeping its energy and forgetting which was which; the line of response is symmetric;
7   positronium's photon count reads the bit of the whole; a bound electron never rests; charge
8   conservation forbids its decay; spin returns to -1 at 2m and to 1 at 4m; a round electron is
9   the fixed set of time reversal. Finite models; every physical number enters as a citation.
10 -/
11 namespace SPHYS.Electron
12
13 /-- A lepton state: charge in units of e, mass in keV, twice the spin projection. -/
14 structure St where
15   q : Int
16   m : Nat
17   s : Int
18   deriving DecidableEq, Repr
19
20 def conj (p : St) : St := (-p.q, p.m, p.s)
21 def neutral (p : St) : Prop := p.q = 0
22 instance (p : St) : Decidable (neutral p) := inferInstanceAs (Decidable (p.q = 0))
23
24 def electron : St := (-1, 511, 1)
25 def positron : St := (1, 511, 1)
26
27 theorem conj_involution (p : St) : conj (conj p) = p := by
28   cases p with
29   | mk q m s => show ((-(-q), m, s) : St) = (q, m, s); rw [Int.neg_neg]
30
31 /-- The seat: a state is its own conjugate exactly when it is neutral. -/
32 theorem self_conjugate_iff_neutral (p : St) : conj p = p ↔ neutral p := by
33   cases p with
34   | mk q m s =>
35     show ((-q, m, s) : St) = (q, m, s) ↔ q = 0
36   constructor
37   · intro h
38     have h1 : -q = q := congrArg St.q h
39     omega
40   · intro h
41     subst h
42     rfl
43
44 /-- The positron is the electron's partner: distinct, and both off the neutral locus. -/
45 theorem positron_is_the_partner : conj electron = positron := by decide
46 theorem the_pair : conj electron ≠ electron ∧ ¬ neutral electron ∧ ¬ neutral positron := by decide
47
48 /-- The pair is exact: conjugation keeps mass and spin and flips only the sign of the charge. -/
49 theorem pair_differs_in_one_sign :
50   electron.m = positron.m ∧ electron.s = positron.s ∧ electron.q = -positron.q := by decide
51
52 /-- Gravity's record is the mass: two different states, one record. -/
53 def massRecord (p : St) : Nat := p.m
54 theorem two_worlds_one_record : massRecord electron = massRecord positron ∧ electron ≠ positron := by
55   decide
56
57 /-- No function of the mass record returns the charge. -/
58 theorem no_reading_of_mass_returns_charge (g : Nat → Int) :
59   ¬ (g (massRecord electron) = electron.q ∧ g (massRecord positron) = positron.q) := by
60   intro (h1, h2)
61   have e1 : g 511 = -1 := h1
62   have e2 : g 511 = 1 := h2
63   omega
64
65 /-- Annihilation as registration: the pair lands on the neutral locus, keeps its energy (the
66   height) and forgets which constituent was which (the side). -/
67 structure Light where
68   q : Int
69   e : Nat
70   deriving DecidableEq, Repr
71 def annihilate (a b : St) : Light := (a.q + b.q, a.m + b.m)
72 theorem annihilation_lands_neutral : (annihilate electron positron).q = 0 := by decide
73 theorem annihilation_keeps_energy : (annihilate electron positron).e = 1022 := by decide
74 theorem annihilation_forgets_side (a b : St) : annihilate a b = annihilate b a := by
75   unfold annihilate
76   rw [Int.add_comm a.q b.q, Nat.add_comm a.m b.m]
77
78 /-- The line of response: two back-to-back photons define one unordered line. -/
79 def online (p x : Int × Int) : Prop := x = p ∨ x = (-p.1, -p.2)
80 theorem line_of_response_symmetric (p x : Int × Int) : online p x ↔ online (-p.1, -p.2) x := by
81   unfold online
82   constructor
83   · intro h
84     cases h with
85     | inl h => right; rw [h, Int.neg_neg, Int.neg_neg]
86     | inr h => left; exact h
87   · intro h
88     cases h with
89     | inl h => right; exact h
90     | inr h => left; rw [h, Int.neg_neg, Int.neg_neg]
91
92 /-- Positronium: the C-parity of the whole is (-1)^(L+S) and n photons carry (-1)^n, so the
93   photon count reads the bit of the whole and never the identity of its parts. -/
94 def sgn (n : Nat) : Int := if n % 2 = 0 then 1 else -1
95 def cParity (L S : Nat) : Int := sgn (L + S)
96 theorem para_two_photons : sgn 2 = cParity 0 0 := by decide
97 theorem ortho_three_photons : sgn 3 = cParity 0 1 := by decide
98 theorem ortho_not_two : sgn 2 ≠ cParity 0 1 := by decide
99
100 /-- The floor: by the virial theorem the kinetic energy of a bound Coulomb state is minus its
101   total energy, so a bound electron never rests. -/
102 theorem virial (K : Int) : (-2 * K) + K = -K := by omega
103 theorem bound_never_rests (E : Int) (hE : E < 0) : 0 < -E := by omega
104
105 /-- Stability: every state lighter than the electron is neutral, so no decay product carries
106   its charge. -/
107 def total : List Int → Int

```

```

108 | [] => 0
109 | a :: t => a + total t
110 theorem total_zero (l : List Int) (h : ∀ q ∈ l, q = 0) : total l = 0 := by
111   induction l with
112   | nil => rfl
113   | cons a t ih =>
114     have ha : a = 0 := h a (List.Mem.head t)
115     have ht : total t = 0 := ih (fun q hq => h q (List.Mem.tail a hq))
116     show a + total t = 0
117     rw [ha, ht]
118     rfl
119 theorem decay_forbidden (l : List Int) (h : ∀ q ∈ l, q = 0) : total l ≠ -1 := by
120   rw [total_zero l h]
121   decide
122
123 /- Spin: the unit quaternion i is a half-turn; twice is a full turn and gives -1; four times
124   returns. Three half-turns about three perpendicular axes give the Return, ijk = -1. -/
125 structure Q where
126   r : Int
127   i : Int
128   j : Int
129   k : Int
130   deriving DecidableEq, Repr
131 def qmul (a b : Q) : Q :=
132   (a.r*b.r - a.i*b.i - a.j*b.j - a.k*b.k, a.r*b.i + a.i*b.r + a.j*b.k - a.k*b.j,
133    a.r*b.j - a.i*b.k + a.j*b.r + a.k*b.i, a.r*b.k + a.i*b.j - a.j*b.i + a.k*b.r)
134 def qi : Q := (0,1,0,0)
135 def qj : Q := (0,0,1,0)
136 def qk : Q := (0,0,0,1)
137 theorem spin_two_pi : qmul qi qi = (-1,0,0,0) := by decide
138 theorem spin_four_pi : qmul (qmul qi qi) (qmul qi qi) = (1,0,0,0) := by decide
139 theorem the_return : qmul (qmul qi qj) qk = (-1,0,0,0) := by decide
140
141 /- The round electron: a dipole moment is odd under time reversal, and the fixed set of time
142   reversal is the zero dipole. -/
143 theorem round_iff_T_fixed (d : Int) : -d = d ↔ d = 0 := by
144   constructor
145   · intro h
146   · omega
147   · intro h
148   · subst h
149   · rfl
150
151 /- The seat, modelled on the critical strip: a point is its doubled real part and its height, and
152   the fold s ↦ 1 - s sends the doubled real part x to 2 - x and keeps the height. -/
153 structure Strip where
154   x : Int
155   t : Nat
156   deriving DecidableEq, Repr
157 def fold (z : Strip) : Strip := (2 - z.x, z.t)
158 def criticalLine (z : Strip) : Prop := z.x = 1
159 theorem fold_involution (z : Strip) : fold (fold z) = z := by
160   cases z with
161   | mk x t =>
162     show Strip.mk (2 - (2 - x)) t = Strip.mk x t
163     congr 1
164     omega
165 theorem fold_fixed_iff (z : Strip) : fold z = z ↔ criticalLine z := by
166   cases z with
167   | mk x t =>
168     show Strip.mk (2 - x) t = Strip.mk x t ↔ x = 1
169     constructor
170     · intro h
171       have hx := congrArg Strip.x h
172       change 2 - x = x at hx
173       omega
174     · intro h
175       subst h
176       rfl
177 /- The explicit equivariant map: charge q goes to doubled real part 1 + q, mass to height. -/
178 def toStrip (p : St) : Strip := (1 + p.q, p.m)
179 theorem toStrip_equivariant (p : St) : toStrip (conj p) = fold (toStrip p) := by
180   show Strip.mk (1 + -p.q) p.m = Strip.mk (2 - (1 + p.q)) p.m
181   congr 1
182   omega
183 /- The neutral states, the fixed set of charge conjugation, land on the critical line. -/
184 theorem neutral_lands_on_line (p : St) (h : neutral p) : criticalLine (toStrip p) := by
185   have hq : p.q = 0 := h
186   show 1 + p.q = 1
187   omega
188 /- A charged pair lands on the two edges of the strip, mirror images under the fold: charge -1 at
189   real part 0, charge +1 at real part 1. -/
190 theorem pair_on_the_edges (p : St) (h : p.q = -1) :
191   (toStrip p).x = 0 ∧ (toStrip (conj p)).x = 2 ∧ toStrip (conj p) = fold (toStrip p) := by
192   refine (?, ?)
193   · show 1 + p.q = 0
194     omega
195   · show 1 + -p.q = 2
196     omega
197
198 end SPHYS.Electron
199 #print axioms SPHYS.Electron.conj_involution
200 #print axioms SPHYS.Electron.self_conjugate_iff_neutral
201 #print axioms SPHYS.Electron.positron_is_the_partner
202 #print axioms SPHYS.Electron.the_pair
203 #print axioms SPHYS.Electron.pair_differs_in_one_sign
204 #print axioms SPHYS.Electron.two_worlds_one_record
205 #print axioms SPHYS.Electron.no_reading_of_mass_returns_charge
206 #print axioms SPHYS.Electron.annihilation_lands_neutral
207 #print axioms SPHYS.Electron.annihilation_keeps_energy
208 #print axioms SPHYS.Electron.annihilation_forgets_side
209 #print axioms SPHYS.Electron.line_of_response_symmetric
210 #print axioms SPHYS.Electron.para_two_photons
211 #print axioms SPHYS.Electron.ortho_three_photons
212 #print axioms SPHYS.Electron.ortho_not_two
213 #print axioms SPHYS.Electron.bound_never_rests
214 #print axioms SPHYS.Electron.decay_forbidden
215 #print axioms SPHYS.Electron.spin_two_pi
216 #print axioms SPHYS.Electron.spin_four_pi
217 #print axioms SPHYS.Electron.the_return
218 #print axioms SPHYS.Electron.round_iff_T_fixed

```

```

219 #print axioms SPHYS.Electron.fold_involution
220 #print axioms SPHYS.Electron.fold_fixed_iff
221 #print axioms SPHYS.Electron.toStrip_equivariant
222 #print axioms SPHYS.Electron.neutral_lands_on_line
223 #print axioms SPHYS.Electron.pair_on_the_edges

```

14. APPENDIX B: ELECTRON_TWIN.F90 AND ITS RUN

SHA-256 ac53e5dc562c3d7cf7ae9daf7037a300592bb606d212d3af7f744cfb177b6e31, standard Fortran 2018, zero warnings under gfortran 13.3.0 -std=f2018 -O2 -fno-fast-math -ffp-contract=off -Wall -Wextra. Seventeen checks, zero failures; every figure computed in this run from the cited constants, the cited bounds entered as data and labelled so.

```

1 ! Electron_Twin.f90 · the executable twin of "The Electron Is the Seat" · Fortran 2018.
2 ! Every figure is computed in this run from cited constants (CODATA 2018 unless stated); an
3 ! oracle stops the program on any failure. Cited bounds enter as data and are labelled so.
4 program electron_twin
5   implicit none
6   integer, parameter :: dp = selected_real_kind(15, 307)
7   real(dp), parameter :: PI = 3.14159265358979323846_dp
8   real(dp), parameter :: MEC2 = 0.51099895000e6_dp ! eV, CODATA 2018
9   real(dp), parameter :: AINVRB = 137.035999084_dp ! CODATA 2018
10  real(dp), parameter :: AINVRB = 137.035999206_dp ! Rb recoil, Morel et al. 2020
11  real(dp), parameter :: AINVCB = 137.035999046_dp ! Cs recoil, Parker et al. 2018
12  real(dp), parameter :: HBAR = 6.582119569e-16_dp ! eV s
13  real(dp), parameter :: RYD = 13.605693122994_dp ! eV, R_inf h c, CODATA 2018
14  real(dp), parameter :: AE_EXP = 0.00115965218059_dp ! Fan et al. 2023
15  real(dp), parameter :: TAU_P = 125.14e-12_dp ! s, para-Ps, Al-Ramadhan & Gidley 1994
16  real(dp), parameter :: TAU_0 = 142.05e-9_dp ! s, ortho-Ps, Vallery et al. 2003
17  real(dp), parameter :: DE_BOUND = 4.1e-30_dp ! e cm, JILA 2023 (cited bound)
18  real(dp), parameter :: LAMBDA_C = 3.8615926796e-11_dp ! cm, reduced Compton wavelength
19  real(dp), parameter :: HBARC = 197.326980466_dp ! eV fm
20  real(dp), parameter :: A0 = 52917.7210903_dp ! fm, Bohr radius
21  real(dp), parameter :: AG = 0.75_dp, AG_S1 = 0.13_dp, AG_S2 = 0.16_dp ! ALPHA-g 2023 (cited)
22  real(dp) :: a, x, egam, e_pair, ryd_calc, k_kin, v_pot, ae_uni, ae_sch, g0p, g0o, tp, to
23  real(dp) :: z_grav, round, dae, m_line, m_bent, tension
24  complex(dp) :: r2(2,2), r4(2,2)
25  real(dp) :: err2, err4
26  integer :: nchk, nfail
27  nchk = 0; nfail = 0
28  ! --- annihilation: the pair lands on light, keeping its energy, forgetting which was which
29  e_pair = 2.0_dp * MEC2
30  egam = e_pair / 2.0_dp
31  call check('annihilation at rest: total energy 2 m c^2 = 1.022 MeV', abs(e_pair - 1.0219979e6_dp) < 1.0_dp)
32  ! invariant mass of two photons at opening angle th: M^2 = 2 E1 E2 (1 - cos th)
33  m_line = sqrt(2.0_dp*egam*egam*(1.0_dp - cos(PI)))
34  m_bent = sqrt(2.0_dp*egam*egam*(1.0_dp - cos(0.9_dp*PI)))
35  call check('back to back, the two photons carry the pair's whole rest energy', abs(m_line/e_pair - 1.0_dp) < 1.0e-12_dp)
36  call check('any other opening angle carries less: the record is forced onto a line', m_bent < e_pair)
37  ! --- the floor: the hydrogen ground state, kinetic energy by the virial theorem
38  a = 1.0_dp / AINVRB
39  ryd_calc = MEC2 * a*a / 2.0_dp
40  ! kinetic and potential energy of the ground state from the Bohr radius and hbar c, independently
41  k_kin = HBARC**2 / (2.0_dp * MEC2 * A0**2)
42  v_pot = -a * HBARC / A0
43  call check('Rydberg energy m c^2 alpha^2 / 2 reproduces R_inf h c to 1e-9', abs(ryd_calc/RYD - 1.0_dp) < 1.0e-9_dp)
44  call check('virial theorem from independent constants: 2T + V = 0 to 1e-9', abs((2.0_dp*k_kin + v_pot)/k_kin) < 1.0e-9_dp)
45  call check('total energy T + V equals minus the Rydberg energy to 1e-9', abs((k_kin + v_pot)/RYD + 1.0_dp) < 1.0e-9_dp)
46  call check('the bound electron never rests: kinetic energy > 0', k_kin > 0.0_dp)
47  ! --- positronium: the photon count reads the bit of the whole; lifetimes with one-loop terms
48  x = a / PI
49  g0p = a**5 * MEC2 / (2.0_dp * HBAR)
50  g0o = 2.0_dp*(PI*PI - 9.0_dp)/(9.0_dp*PI) * a**6 * MEC2 / HBAR
51  tp = 1.0_dp / (g0p * (1.0_dp - x*(5.0_dp - PI*PI/4.0_dp)))
52  to = 1.0_dp / (g0o * (1.0_dp - 10.286606_dp * x))
53  call check('para-positronium (C even, two photons): lifetime within 0.5% of measured', abs(tp/TAU_P - 1.0_dp) < 5.0e-3_dp)
54  call check('ortho-positronium (C odd, three photons): lifetime within 0.5% of measured', abs(to/TAU_0 - 1.0_dp) < 5.0e-3_dp)
55  call check('the whole bit sets a thousandfold lifetime ratio', to/tp > 1000.0_dp)
56  ! --- spin: a full turn gives -1, two full turns return (SU(2), rotation about z)
57  r2 = rotz(2.0_dp*PI); r4 = rotz(4.0_dp*PI)
58  err2 = abs(r2(1,1) + 1.0_dp) + abs(r2(2,2) + 1.0_dp) + abs(r2(1,2)) + abs(r2(2,1))
59  err4 = abs(r4(1,1) - 1.0_dp) + abs(r4(2,2) - 1.0_dp) + abs(r4(1,2)) + abs(r4(2,1))
60  call check('spin: a rotation by 2 pi is -1 on the spinor', err2 < 1.0e-12_dp)
61  call check('spin: a rotation by 4 pi returns the spinor', err4 < 1.0e-12_dp)
62  ! --- g - 2: the universal QED series to five loops, alpha from rubidium recoil
63  a = 1.0_dp / AINVRB; x = a / PI
64  ae_sch = x / 2.0_dp
65  ae_uni = 0.5_dp*x - 0.328478965579193_dp*x**2 + 1.181241456587_dp*x**3 &
66  - 1.912245764926_dp*x**4 + 6.737_dp*x**5
67  dae = ae_uni - AE_EXP
68  call check('Schwinger term alpha/2pi fixes a_e to three figures', abs(ae_sch - AE_EXP) < 2.0e-6_dp)
69  call check('five-loop universal series meets experiment to 1e-11', abs(dae) < 1.0e-11_dp)
70  ! --- cited bounds, entered as data
71  round = DE_BOUND / LAMBDA_C
72  call check('cited: EDM bound below 1e-18 of the electron's own length scale', round < 1.0e-18_dp)
73  z_grav = abs(1.0_dp - AG) / sqrt(AG_S1**2 + AG_S2**2)
74  call check('cited: antihydrogen falls as matter does, within two standard deviations', z_grav < 2.0_dp)
75  tension = abs(AINVRB - AINVCB) / sqrt(0.000000011_dp**2 + 0.000000027_dp**2)
76  call check('cited: the rubidium and caesium values of alpha differ by more than five sigma (open)', tension > 5.0_dp)
77  write(*, '(a,f14.3,a)') ' annihilation energy = ', e_pair/1.0e3_dp, ' keV'
78  write(*, '(a,f14.9,a)') ' Rydberg energy (computed) = ', ryd_calc, ' eV'
79  write(*, '(a,f14.9,a,f14.9,a)') ' ground state T, V = ', k_kin, ' eV', v_pot, ' eV'
80  write(*, '(a,f10.3,a,f10.3,a)') ' para-Ps lifetime = ', tp*1.0e12_dp, ' ps (measured)', TAU_P*1.0e12_dp, ' ps)'
81  write(*, '(a,f10.3,a,f10.3,a)') ' ortho-Ps lifetime = ', to*1.0e9_dp, ' ns (measured)', TAU_0*1.0e9_dp, ' ns)'
82  write(*, '(a,f10.1)') ' lifetime ratio ortho/para = ', to/tp
83  write(*, '(a,es10.3)') ' spinor error at 2 pi = ', err2
84  write(*, '(a,f18.15)') ' a_e, universal five-loop = ', ae_uni
85  write(*, '(a,f18.15)') ' a_e, measured (Fan 2023) = ', AE_EXP
86  write(*, '(a,es10.3)') ' difference = ', dae
87  write(*, '(a,es10.3)') ' EDM bound / lambda_C = ', round
88  write(*, '(a,f6.2)') ' ALPHA-g deviation (sigma) = ', z_grav
89  write(*, '(a,f6.2)') ' alpha tension Rb-Cs (sig) = ', tension
90  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', nchk, ', "failures":', nfail, '}'
91  if (nfail > 0) error stop 1
92 contains

```

```

93 function rotz(theta) result(r)
94   real(dp), intent(in) :: theta
95   complex(dp) :: r(2,2)
96   r = (0.0_dp, 0.0_dp)
97   r(1,1) = exp(cmplx(0.0_dp, -theta/2.0_dp, dp))
98   r(2,2) = exp(cmplx(0.0_dp, theta/2.0_dp, dp))
99 end function rotz
100 subroutine check(label, cond)
101   character(*), intent(in) :: label
102   logical, intent(in) :: cond
103   nchk = nchk + 1
104   if (.not. cond) then
105     nfail = nfail + 1; write(*,'(a,a)') ' FAIL ', label
106   end if
107 end subroutine check
108 end program electron_twin

```

The run, verbatim.

```

1 annihilation energy = 1021.998 keV
2 Rydberg energy (computed) = 13.605693123 eV
3 ground state T, V = 13.605693115 eV, -27.211386238 eV
4 para-Ps lifetime = 125.231 ps (measured 125.140 ps)
5 ortho-Ps lifetime = 142.068 ns (measured 142.050 ns)
6 lifetime ratio ortho/para = 1134.5
7 spinor error at 2 pi = 2.449E-16
8 a_e, universal five-loop = 0.001159652175783
9 a_e, measured (Fan 2023) = 0.001159652180590
10 difference = -4.807E-12
11 EDM bound / lambda_C = 1.062E-19
12 ALPHA-g deviation (sigma) = 1.21
13 alpha tension Rb-Cs (sig) = 5.49
14 BATTERY-JSON: {"checks":17,"failures":0}

```